Total asphericity and exact abelian colimits

TOTEX2: INCOMPLETE interruption record

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13 November 2026

Abstract

This is an incomplete research record prepared after an interruption. The main implication and the universal AKR persistence assertion remain unresolved. The verified fragments are preserved here for continued work. We distinguish proved statements from the unresolved implication from total asphericity to exact abelian colimits. No priority is claimed. All categories are small; all set and module diagrams are covariant unless explicitly stated otherwise.

Contents

A The main implication and coefficient detection **1**

B Triples and the five algebraic leads **3**
B.1 The exact triple question and the checked dependencies 3
B.2 The intersection and one-relation tests 4
B.3 Set-colimit equality, abelian-colimit equality, and the mixed cycle 6

C The Adámek–Koubek–Reiterman construction and finite tests **9**
C.1 Finite tests and comparison of links 9
C.2 The interval-type candidate and necessary conditions 11
C.3 The checked expansion invariant and a sharper obstruction 13
C.4 Status of the all-stage homology assertion 15
C.5 The homogeneous Ramsey candidates 15

D Class inclusions, manuscript corrections, and sources **17**
D.1 The class diagram and the manuscript corrections 17
D.2 Primary-source audit and main-question search 21

A The main implication and coefficient detection

Definition A.1 (Known: conventions)**.** For a small category A we put $BA = |NA|$. For $H : A \to \mathrm{Set}$, the category $\mathrm{El}_A H$ has objects (a, h) and arrows $f : (a, h) \to (b, k)$ satisfying $H(f)h = k$. We call A total when it is nonempty and connected and every component of every $\mathrm{El}_A H$ is weakly contractible. We call A exact when $\mathrm{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ is exact. We write $P_x = \mathbb{Z}[A(x, -)]$ and $T_i(M) = L_i \,\mathrm{colim}_A M$.

Unresolved statement A.2 (Main question)**.** Does every total small category have exact abelian colimits? No proof or counterexample is asserted at this checkpoint.

Theorem A.3 (Proved: detection by all prime fields)**.** *Let A be any small category, possibly empty or disconnected. The following are equivalent:*

(i) $\operatorname{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ *is exact.*

(ii) Colimits over A are exact for $\mathbb{Q}$-vector-space diagrams and for $\mathbb{F}_p$-vector-space diagrams for every prime p.

Consequently the conjunction of all coefficient implications requested in Part A.2 is equivalent to Part A.1. This is a statement about all primes together; it makes no assertion that finitely many field tests suffice.

Proof. The underlying abelian group of a colimit of R-module diagrams is its abelian-group colimit, with its uniquely induced R-action. Indeed, the usual direct-sum presentation and its relation subgroup are R-linear. Thus (i) implies (ii).

Assume (ii). For a covariant abelian diagram M, let $C_\bullet(A, M)$ be its bar complex, with

$$C_n(A, M) = \bigoplus_{a_0 \to \cdots \to a_n} M(a_0).$$

The first face applies the transition map of M, the intermediate faces compose arrows, and the last face omits the last arrow; degeneracies insert identities. Its homology is $T_n(M)$. This statement applies even when the groups $M(a)$ are not projective: an objectwise projective resolution in Ab^A, followed by the bar construction, gives a first-quadrant double complex; direct sums are exact in the resolution direction, and the augmented bar complex contracts on each representable projective in the other direction. The two filtrations give the asserted comparison.

First suppose every $M(a)$ is torsionfree. The complex $C_\bullet = C_\bullet(A, M)$ consists of flat $\mathbb{Z}$-modules, and tensoring it with a field k gives literally $C_\bullet(A, M \otimes_{\mathbb{Z}} k)$. By (ii), its positive homology vanishes for $k = \mathbb{Q}$ and for every $k = \mathbb{F}_p$. Flat base change to $\mathbb{Q}$ gives

$$H_i(C_\bullet) \otimes \mathbb{Q} = 0 \qquad (i > 0),$$

so each $H_i(C_\bullet)$ is torsion. For a complex of flat $\mathbb{Z}$-modules the universal-coefficient sequence is

$$0 \longrightarrow H_j(C_\bullet) \otimes \mathbb{F}_p \longrightarrow H_j(C_\bullet \otimes \mathbb{F}_p) \longrightarrow \operatorname{Tor}_1^{\mathbb{Z}}(H_{j-1}(C_\bullet), \mathbb{F}_p) \longrightarrow 0.$$

For completeness, this follows by tensoring the cycle and boundary short exact sequences; their boundary groups are torsionfree subgroups of flat abelian groups and hence flat. Taking $j = i + 1$ shows that $H_i(C_\bullet)[p] = 0$ for every prime p. A nonzero torsion abelian group contains an element of prime order, so $H_i(C_\bullet) = 0$. Thus every objectwise torsionfree diagram is colimit-acyclic.

Next suppose M is objectwise torsion. It decomposes naturally as the direct sum of its p-primary subfunctors. A p-primary diagram is the filtered union of the subfunctors $M[p^n]$. Each quotient $M[p^n]/M[p^{n-1}]$ is killed by p, and therefore is a diagram of $\mathbb{F}_p$-vector spaces. Its integral bar complex is the underlying abelian-group complex of its $\mathbb{F}_p$ bar complex, so (ii) makes it acyclic. Induction on n using the long exact sequence of derived colimits proves acyclicity of $M[p^n]$. The bar complex and homology commute with direct sums and filtered colimits of abelian groups, proving acyclicity of M.

For arbitrary M, its torsion subgroups form a subfunctor M_{tor}. The quotient M/M_{tor} is objectwise torsionfree. Both ends of the short exact sequence

$$0 \longrightarrow M_{\mathrm{tor}} \longrightarrow M \longrightarrow M/M_{\mathrm{tor}} \longrightarrow 0$$

are acyclic by the preceding paragraphs, hence so is M. In particular $T_1(M) = 0$ for every M. Since Ab^A has enough projectives and colimit is right exact, this is exactness of colim_A. Empty and disconnected categories cause no exception in the argument. □

Proposition A.4 (Proved: the resolving closure and its tensor operation)**.** *For a total category A, let $\mathcal{R}_A$ be the smallest full class of objects of Ab^A containing all $\mathbb{Z}[H]$, $H : A \to \mathrm{Set}$, and closed under extensions, kernels of epimorphisms between its objects, direct sums, direct summands, and filtered colimits. Every object of $\mathcal{R}_A$ is colimit-acyclic. Moreover, for every $H : A \to \mathrm{Set}$, the functor $M \mapsto M \otimes_{\mathbb{Z}} \mathbb{Z}[H]$ preserves $\mathcal{R}_A$. Adding the tensor operation to the definition therefore does not enlarge this class.*

Proof. The bar complex of $\mathbb{Z}[H]$ is the integral nerve-chain complex of $\mathrm{El}_A H$, so totality proves the initial vanishing. Long exact sequences give the extension and kernel closures, and the bar complex gives the other three closures. For fixed H, pointwise tensoring with $\mathbb{Z}[H]$ is exact, preserves direct sums, retracts, and filtered colimits, and sends $\mathbb{Z}[G]$ to $\mathbb{Z}[G \times H]$. The class of all M for which $M \otimes \mathbb{Z}[H] \in \mathcal{R}_A$ consequently contains the generators and has every closure used to define $\mathcal{R}_A$. Minimality gives the assertion. We have not asserted tensor stability for the larger class of all colimit-acyclic diagrams. □

Remark A.5 (Unresolved: the membership step)**.** For an abelian diagram M, the augmentation ideal $I(M) = \ker(\mathbb{Z}[UM] \to \mathbb{Z})$ and all its tensor powers lie in $\mathcal{R}_A$. The natural epimorphism $I(M) \to M$, $[m] - [0] \mapsto m$, has kernel $I(M)^2$, where multiplication is in the group ring of the additive group $M(a)$ at each object. Although $I(M)^2$ is the image of $I(M) \otimes I(M) \to I(M)$, images are not among the proved closure operations. No acyclicity of that image, or membership of the column modules in $\mathcal{R}_A$, is inferred here.

B Triples and the five algebraic leads

B.1 The exact triple question and the checked dependencies

Unresolved statement B.1 (Part B.1: not settled in this record)**.** We have not proved or refuted that every total category supplies a simultaneous affine equalizer for every three ordinary parallel arrows. Precisely, for $a_0, a_1, a_2 : p \to q$, the unproved vanishing is

$$T_1\Big(\mathrm{coker}\big(P_q \longrightarrow P_p^2, t \longmapsto (t(a_1 - a_0), t(a_2 - a_0))\big)\Big) = 0.$$

The propositions below identify this group and check all five proposed algebraic leads. They do not assert its vanishing.

We first record the dependencies actually checked. We read TOTEX A.1–A.5, B.1–B.7 and D.13, the corresponding proofs in TOTASPH C.3–C.6 and C.11–C.13, and the affine-criterion proof in TOTASPH D. We also opened the supplied Isbell and Mitchell (1976), in particular Lemma 2.2 and Proposition 2.3 on p. 293, Theorem 3.1 on pp. 294–295, Proposition 3.3 on p. 295 and the coefficient remark on p. 296.

For R1 the step used here is literal: the degree-n generators of the bar complex of $\mathbb{Z}[H]$ consist of a composable n-chain in A and an element of H at its source. These are precisely the nerve simplices of $\mathrm{El}_A(H)$, with the same faces. Thus totality makes $T_i(\mathbb{Z}[H])$ vanish for $i > 0$, without assuming that H is connected, nonempty, or has injective transition maps. The source's dependent-sum construction correctly transfers actions on an element component to actions on A; its cyclic-cover argument applies to unrestricted actions. We do not replace $\mathbb{Z}[H]$ by an arbitrary module.

For R2, Isbell–Mitchell's Theorem 3.1 equates exactness with filtered components of $\mathrm{Aff}_{\mathbb{Z}}(A)$, and Proposition 3.3 equates this with I and III. The finite-support step in the necessity proof really uses common successors to put a finite sum at one target. The sufficient direction follows directly because affine extension preserves the colimit: for $\sum n_i = 1$,

$$x - \Big(\sum_i n_i f_i\Big)x = \sum_i n_i (x - f_i x).$$

For R4, the common-successor argument applies also to the total element category p/A, and therefore yields I. The pair presentation and H_1 vanishing yield an affine equalizer after finitely many augmentation witnesses have been moved to one target. We give below a direct proof of the instance of Proposition 2.3 that we use. For R7, we have checked the two short exact sequences involving the image of the relation map and the fact that the induced map on representable colimits is zero. In particular its row kernel in P_q^2 is kept distinct from the column kernel in P_q.

All modules in this section are covariant functors $A \to \mathrm{Ab}$ and every displayed module map is a natural $\mathbb{Z}$-linear transformation. We use the canonical identification $\operatorname{colim}_A P_x = \mathbb{Z}$ by augmentation. For a submodule $E \subset P_x$, $\varepsilon(E)$ means the image of $\operatorname{colim} E \to \mathbb{Z}$, not a pointwise union. When I holds, every finite collection of objects in one component has a common successor: completion of spans shortens a finite connecting zigzag, and induction handles finitely many objects.

B.2 The intersection and one-relation tests

Theorem B.2 (Proved: Lead 1)**.** *Let A be total, let $a_0, a_1, a_2 : p \to q$, and put*

$$z_i = a_i - a_0, \quad K_i = \ker(P_q \xrightarrow{z_i} P_p), \quad L = K_1 \cap K_2, \quad S = K_1 + K_2, \quad C = P_q/S.$$

Then the surjection of R7 is canonically an isomorphism

$$T_1(C) \cong \mathbb{Z}/\varepsilon(L).$$

Moreover, with $J_i = P_q z_i \subset P_p$ and $D = \operatorname{im}(P_q \to J_1 \oplus J_2,\ t \mapsto (tz_1, tz_2))$,

$$D \cong P_q/L, \qquad C \cong (J_1 \oplus J_2)/D, \qquad \operatorname{colim} J_i = 0, \quad T_j(J_i) = 0 \quad (j > 0).$$

Proof. Define $H_i = A(p, -)/(ca_0 \sim ca_i)$. Free abelianization preserves this set coequalizer, giving

$$0 \longrightarrow J_i \longrightarrow P_p \longrightarrow \mathbb{Z}[H_i] \longrightarrow 0.$$

The last module is acyclic by totality. Its colimit is $\mathbb{Z}$: every element of H_i is reached from the distinguished element at p, so its category of elements is connected. The induced map $\operatorname{colim} P_p \to \operatorname{colim} \mathbb{Z}[H_i]$ is the identity of $\mathbb{Z}$. The long exact sequence therefore gives $\operatorname{colim} J_i = 0$ and $T_j(J_i) = 0$ for $j > 0$. Applying it to $0 \to K_i \to P_q \to J_i \to 0$ proves that the augmentation $\operatorname{colim} K_i \to \mathbb{Z}$ is an isomorphism and that each K_i is acyclic.

The pointwise exact sequence

$$0 \longrightarrow L \xrightarrow{l \mapsto (l,-l)} K_1 \oplus K_2 \xrightarrow{(u,v) \mapsto u+v} S \longrightarrow 0$$

gives

$$\operatorname{colim} S \cong \mathbb{Z}^2/\{(n, -n) : n \in \varepsilon(L)\}.$$

The map to $\operatorname{colim} P_q = \mathbb{Z}$ is induced by $(a, b) \mapsto a + b$. Its kernel is canonically $\mathbb{Z}/\varepsilon(L)$ through $(n, -n) \mapsto n$. Since P_q is projective, $0 \to S \to P_q \to C \to 0$ identifies this kernel with $T_1(C)$. This is precisely the map constructed in R7.

The map defining D has kernel L, so $D \cong P_q/L$. The other isomorphism is explicit:

$$P_q/(K_1 + K_2) \longrightarrow (J_1 \oplus J_2)/D, \qquad [t] \longmapsto [(tz_1, 0)].$$

It is surjective because subtracting (vz_1, vz_2) from (uz_1, vz_2) leaves $((u - v)z_1, 0)$. Its kernel is $K_1 + K_2$: the equality $(tz_1, 0) = (wz_1, wz_2)$ says $w \in K_2$ and $t - w \in K_1$, and conversely these conditions give that equality. All maps are natural in the target object. Finally $0 \to D \to J_1 \oplus J_2 \to C \to 0$ gives the alternative identification $T_1(C) \cong \operatorname{colim} D = \mathbb{Z}/\varepsilon(L)$. □

Theorem B.3 (Proved: Lead 2, including all component and empty cases)**.** *Let A be a small category satisfying I. Its abelian colimit functor is exact if and only if, for all $p, r \in A$ and every $z \in \mathbb{Z}[A(p,r)]$ with $\operatorname{aug} z = 0$, there are $s \in A$ and $t \in \mathbb{Z}[A(r,s)]$ such that $\operatorname{aug} t = 1$ and $tz = 0$. Without any hypothesis on A, putting $E = \operatorname{ann}(z)$ gives the canonical formula*

$$T_1(P_p/P_r z) \cong \mathbb{Z}/\varepsilon(E).$$

For a total category this annihilator property holds in particular for every binomial $z = c - c'$.

Proof. Let $J = P_r z \subset P_p$. From $0 \to E \to P_r \to J \to 0$ we obtain $\operatorname{colim} J = \mathbb{Z}/\varepsilon(E)$. The map $\operatorname{colim} J \to \operatorname{colim} P_p = \mathbb{Z}$ is zero: its composite with the epimorphism $\mathbb{Z} = \operatorname{colim} P_r \to \operatorname{colim} J$ is multiplication by $\operatorname{aug} z = 0$. The sequence $0 \to J \to P_p \to P_p/J \to 0$, with projective middle term, proves the displayed formula.

If colimits are exact, this Tor group vanishes, so $\varepsilon(E) = \mathbb{Z}$. Represent 1 by a finite sum of elements of E at possibly different objects. All those objects lie in the component of r, and I gives a common successor. Transporting and adding the representatives gives one t with augmentation 1 and $tz = 0$.

Conversely, given parallel affine arrows $\alpha, \beta : p \to r$, apply the assumed property to $z = \alpha - \beta$. It supplies an affine equalizer for them. Condition I gives common successors in each component of A, and $\operatorname{Aff}_{\mathbb{Z}}(A)$ has the same components. Its components are therefore filtered. Theorem 3.1 of Isbell and Mitchell (1976) gives exactness, or one can extend every module affinely and compute its unchanged colimit over those filtered components. The binomial assertion is R4's pair criterion.

If $z = 0$, then $E = P_r$ and id_r is an annihilator, agreeing with the formula. When $A(p,r) = \varnothing$ this is the only z. For the empty category both quantified conditions are vacuous and the zero colimit functor is exact. No global connectedness assumption has been used. □

A theorem for three ordinary arrows would not by itself prove Theorem B.3's annihilator condition: after an affine equalizer is applied to three arrows, later comparisons are generally between affine arrows. We make no induction from ordinary triples to arbitrary finite families.

Theorem B.4 (Proved: Lead 3 without hypotheses on the category)**.** *Let A be any small category, $k \geq 0$ finite, and $a_0, \dots, a_k : p \to q$. For $z_i = a_i - a_0$, define*

$$d^T : P_q \longrightarrow P_p^k, \quad t \longmapsto (tz_1, \dots, tz_k), \qquad L = \ker d^T, \quad Q_a = \operatorname{coker} d^T.$$

There are canonical isomorphisms

$$\begin{aligned} T_1(Q_a) &\cong \mathbb{Z}/\varepsilon(L), \\ T_2(Q_a) &\cong \ker(\operatorname{colim} L \to \mathbb{Z}), \\ T_i(Q_a) &\cong T_{i-2}(L) \qquad (i \geq 3). \end{aligned}$$

If finite sets of objects in the component of q have common successors, the family has a simultaneous affine equalizer if and only if $T_1(Q_a) = 0$. The row map

$$d : P_q^k \longrightarrow P_p, \qquad (t_i) \longmapsto \sum_i t_i z_i$$

has cokernel $\mathbb{Z}[A(p,-)/(ca_0 \sim \cdots \sim ca_k)]$. The module Q_a has generators $x_1, \dots, x_k$ at p and the single module relation $\sum_i z_i x_i = 0$ at q.

Proof. Put $D = \operatorname{im} d^T$. The first short exact sequence $0 \to L \to P_q \to D \to 0$ gives

$$\operatorname{colim} D = \mathbb{Z}/\varepsilon(L), \quad T_1(D) = \ker(\operatorname{colim} L \to \mathbb{Z}), \quad T_j(D) = T_{j-1}(L) \quad (j \geq 2).$$

The map $\operatorname{colim} D \to \operatorname{colim} P_p^k = \mathbb{Z}^k$ is zero because its composite with the epimorphism $\operatorname{colim} P_q \to \operatorname{colim} D$ is the vector of augmentations of the z_i, all zero. The long exact sequence of $0 \to D \to P_p^k \to Q_a \to 0$ gives the three asserted formulas. Canonicity is inherited from the connecting morphisms and the representable augmentations.

An affine equalizer is exactly an element of $L(r)$ with augmentation 1. Such an element gives $\varepsilon(L) = \mathbb{Z}$. Conversely, common successors allow a finite family of representatives witnessing $1 \in \varepsilon(L)$ to be transported and added at one target. This proves the equivalence. For $k = 0$, $P_p^0 = 0$, $L = P_q$ and $Q_a = 0$, so all formulas remain valid.

Objectwise, the free abelian group on a quotient set is the quotient of the original free group by the subgroup generated by differences of identified elements. Applying this to the specified relations gives the cokernel of d. The generators x_i are the identity elements in the ith summands of P_p^k, and $\sum_i z_i x_i$ is an element of $(P_p^k)(q)$. This types the last presentation. Under totality the row cokernel is acyclic by R1; transposing this matrix does not identify its cokernel with Q_a. □

B.3 Set-colimit equality, abelian-colimit equality, and the mixed cycle

Theorem B.5 (Proved: the quantified assertions of Lead 4)**.** *Suppose A is quasi-filtered, $M : A \to \mathrm{Ab}$, and $\alpha, \beta \in M(q)$ have the same class in $\operatorname{colim}_A^{\mathrm{Set}} UM$. Then an affine $t : q \to s$ satisfies $M(t)\alpha = M(t)\beta$.*

More generally, assuming only I on A, exactness of $\operatorname{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ is equivalent to the same conclusion for every M, q, α, β whose classes agree in $\operatorname{colim}_A^{\mathrm{Ab}} M$.

Proof. In the first assertion let X be the connected component subfunctor of UM containing α and β. Condition I on A reduces the finite zigzag in X to arrows $f, g : q \to r$ with $M(f)\alpha = M(g)\beta$, exactly as in Isbell and Mitchell (1976, Lemma 2.2). Condition IV gives an affine $v : r \to s$ with $vf = vg$. Put $t = vf = vg$. Then

$$M(t)\alpha = M(v)M(f)\alpha = M(v)M(g)\beta = M(t)\beta.$$

This directly checks the instance of their Proposition 2.3 that is required.

For necessity of the second assertion, extend M affinely. By exactness and Theorem 3.1, the components of $\mathrm{Aff}_{\mathbb{Z}}(A)$ are filtered. The affine-extension comparison identifies the abelian colimits. A filtered colimit of abelian groups has the same underlying set as the filtered colimit of its underlying sets: a finite collection of representatives can be moved to one object, and equality is then witnessed after one transition. Thus the equal images of α and β are witnessed by a single affine arrow t out of q. Components outside that of q contribute separate direct summands and do not affect this equality.

For sufficiency apply the asserted property to $M = P_p$ and parallel affine $\alpha, \beta : p \to q$. Their images in $\operatorname{colim} P_p = \mathbb{Z}$ are both 1. We obtain an affine equalizer of every affine pair; I then gives filtered components of $\mathrm{Aff}_{\mathbb{Z}}(A)$ and hence exactness. □

Proposition B.6 (Proved qualification of Lead 4; stronger readings refuted)**.** *There is a canonical map*

$$\theta_M : \operatorname*{colim}_A^{\mathrm{Set}} UM \longrightarrow U(\operatorname*{colim}_A^{\mathrm{Ab}} M).$$

For a nonempty connected A with common successors it is surjective. Under I and IV, the main affine-equalization question asks whether all pairs in its fibres admit affine equalizers, since pairs equal already in the source do. This is not an equivalence between exactness and injectivity of θ_M, nor an equivalence between exactness and preservation of finite products by set colimits: both stronger assertions are false.

Proof. Every abelian-colimit element is a finite sum of representatives. Common successors put them at one object, where they can be added, proving surjectivity. The assertion about fibres is exactly Theorem B.5. With multiple components surjectivity must instead be asserted componentwise, since a sum may involve several components.

For the first stronger assertion take $A = \Delta_{\mathrm{inj}}$, the countable category of nonempty finite ordinals and increasing injections, and $M = P_{[0]}$. Its exactness is the checked R3. An element of $M([n])$ is an integer vector; an increasing injection inserts zero coordinates, hence preserves the ordered list of its nonzero coefficients. At [2] the vectors

$$\alpha = (1, 1, -1), \qquad \beta = (1, 0, 0)$$

have different such lists and therefore belong to distinct components of $\mathrm{colim}^{\mathrm{Set}}\, UM$. They both have augmentation 1, so their abelian-colimit images agree. Indeed the explicit affine arrow

$$t = (0, 1, 2) - (0, 1, 3) + (0, 2, 3) : [2] \longrightarrow [3]$$

has augmentation 1 and sends both vectors to $(1, 0, 0, 0)$. The program `code/check_delta_patterns.py` checks this identity and the pattern invariant for all increasing injections into target sizes 3 through 9; the zero-insertion proof applies to every target size.

For the second stronger assertion, the cospan category on $[0], [0]$ in Δ_{inj} has three components, according as the distinguished points satisfy $<$, $=$ or $>$. Increasing injections preserve these relations; each class is connected by its smallest representative. Hence the diagonal is not final, so set colimits do not preserve finite products, although abelian colimits are exact. The informal final sentence of Lead 4 is valid only in its fibrewise affine-equalization formulation, not as either stronger equivalence. □

Theorem B.7 (Proved: Lead 5, with its coefficient-cycle meaning specified)**.** *Fix $a_0, a_1, a_2 : p \to q$, write $z_i = a_i - a_0$, and put*

$$K = \ker(P_q^2 \to P_p,\ (t_1, t_2) \mapsto t_1 z_1 + t_2 z_2), \quad K_i = \mathrm{ann}(z_i), \quad K_3 = \mathrm{ann}(z_2 - z_1).$$

Let $\rho_1, \rho_2, \rho_3 \in P_q(s)$ have augmentation 1 and satisfy $\rho_1 z_1 = 0$, $\rho_2 z_2 = 0$, $\rho_3(z_2 - z_1) = 0$. Set $\kappa = (\rho_1 - \rho_3, \rho_3 - \rho_2)$.

(i) We have $\kappa \in K(s)$ and $\varepsilon^2(\kappa) = 0$. If A is total, its class in $\mathrm{colim}\, K$ is zero.

(ii) If A is connected and satisfies I, $F : A \to \mathrm{Ab}$, and $\xi \in F(s)$, then $[\xi] = 0$ in $\mathrm{colim}\, F$ if and only if there are an arrow $u : s \to y$, finitely many parallel pairs $e_j, e'_j : x_j \rightrightarrows y$, and $k_j \in F(x_j)$ such that

$$F(u)\xi = \sum_j (F(e_j) - F(e'_j))k_j.$$

(iii) If an affine $t : s \to r$ annihilates κ, then the common value $t\rho_1 = t\rho_2 = t\rho_3$ is an affine equalizer of the triple. Such a t exists if the parallel pairs in (ii), applied to $F = K$ and $\xi = \kappa$, have a common affine equalizer. For quasi-filtered A, one term in (ii) suffices.

(iv) The natural map

$$\psi : K_1 \oplus K_2 \oplus K_3 \longrightarrow K, \qquad (k_1, k_2, k_3) \longmapsto (k_1 - k_3, k_2 + k_3)$$

has kernel canonically isomorphic to $L = K_1 \cap K_2$. For total A, putting $C = K/\operatorname{im}\psi$ gives

$$T_1(C) \cong \mathbb{Z}/\varepsilon(L).$$

At each object x, $K(x)$ is the integral cycle group of the bipartite multigraph with vertices $A(q, x) \amalg A(p, x)$ and edges $g \to ga_i$ for $i = 0, 1, 2$; $\operatorname{im}\psi(x)$ is the sum of the cycle groups of the three two-colour subgraphs.

Proof. For (i), direct expansion gives

$$(\rho_1 - \rho_3)z_1 + (\rho_3 - \rho_2)z_2 = \rho_1 z_1 - \rho_2 z_2 + \rho_3(z_2 - z_1) = 0.$$

Both coordinate augmentations vanish. The checked R7 row calculation, applied to total A, makes $\operatorname{colim} K \to \mathbb{Z}^2$ an isomorphism, so $[\kappa] = 0$. The two coordinates are restrictions of the representable augmentations; K has not been identified with L.

For (ii), in fact common successors for finite sets of objects suffice. The colimit is the quotient of $\bigoplus_x F(x)$ by the subgroup generated by $i_v F(f)k - i_w k$ for arrows $f : w \to v$. If $[\xi] = 0$, choose a finite expression

$$i_s \xi = \sum_j (i_{v_j} F(f_j) k_j - i_{w_j} k_j).$$

Absorb integer coefficients and signs into the k_j. Choose a common successor y of s and all v_j, w_j, and one arrow $h_x : x \to y$ for each distinct object appearing. Apply the homomorphism from this finite direct sum to $F(y)$ whose x-component is $F(h_x)$. Then $u = h_s$, $e_j = h_{v_j} f_j$ and $e'_j = h_{w_j}$ give exactly the asserted equation. Conversely every parallel difference has zero colimit class and $[F(u)\xi] = [\xi]$. No transition map of F has been assumed injective.

For (iii), $K(t)\kappa = 0$ gives $t\rho_1 = t\rho_3 = t\rho_2$ in $P_q(r)$. The common value has augmentation 1 and annihilates both z_1 and z_2. If $v : y \to r$ is a common affine equalizer of the pairs from (ii), applying $K(v)$ to that equation gives $K(vu)\kappa = 0$, and vu is affine. With one term, IV supplies v. This last deduction requires the stated pair-equalizer hypothesis; I alone would not supply it.

For (iv), the map lands in K because $k_1 z_1 + k_2 z_2 + k_3(z_2 - z_1) = 0$. Its kernel consists exactly of triples $(l, -l, l)$ with $l \in L$; membership of l in K_3 follows from the two defining annihilations. Thus $l \mapsto (l, -l, l)$ is the specified kernel isomorphism.

Assume totality and put $S = \operatorname{im} \psi$. The pair calculation of Theorem B.2, also applied to (a_1, a_2), makes every K_i acyclic with augmentation colimit $\mathbb{Z}$. The row calculation makes K acyclic with colimit $\mathbb{Z}^2$. Right exactness of $0 \to L \to \bigoplus_{i=1}^3 K_i \to S \to 0$ gives

$$\operatorname{colim} S \cong \mathbb{Z}^3 / \{(n, -n, n) : n \in \varepsilon(L)\}.$$

Its map to $\operatorname{colim} K$ is induced by $f : \mathbb{Z}^3 \to \mathbb{Z}^2$, $f(a, b, c) = (a - c, b + c)$. This map is surjective and has kernel $\{(n, -n, n) : n \in \mathbb{Z}\}$. Since K is acyclic, the long exact sequence of $0 \to S \to K \to C \to 0$ gives the claimed isomorphism. Under its connecting map, $n \bmod \varepsilon(L)$ goes to the class of $(n, -n, n)$ in $\operatorname{colim} S$.

For the graph assertion, orient every edge from the $A(q, x)$ side to the $A(p, x)$ side. Its degree-one group is $P_q(x)^3$, and its boundary is

$$(t_0, t_1, t_2) \longmapsto (-t_0 - t_1 - t_2,\ t_0 a_0 + t_1 a_1 + t_2 a_2).$$

The natural isomorphism with its cycle group is

$$K(x) \longrightarrow Z_1(G_x; \mathbb{Z}), \qquad (t_1, t_2) \longmapsto (-t_1 - t_2, t_1, t_2),$$

with inverse taking the last two coordinates. Restricting to colours $0, 1$, to $0, 2$, or to $1, 2$ gives exactly the three summands in the image of ψ. Repeated endpoints retain distinct edges, so the argument includes coincident composites and noninjective transition maps. □

Corollary B.8 (Proved: higher derived groups and the precise mixed-cycle witness)**.** *Under the totality hypotheses of Theorem B.7, its module $C = K/\operatorname{im}\psi$ also satisfies*

$$T_2(C) \cong \ker(\operatorname{colim} L \to \mathbb{Z}), \qquad T_j(C) \cong T_{j-2}(L) \quad (j \geq 3).$$

The class $1 \bmod \varepsilon(L)$ in $T_1(C)$ has connecting image $[\kappa] \in \operatorname{colim}(\operatorname{im}\psi)$. If the triple has no affine equalizer, this is a nonzero class. It is represented by a bar 1-cycle with coefficients in C, not by a nonzero ordinary homology class of BA.

Proof. Write $S = \operatorname{im}\psi$. Acyclicity of $K_1 \oplus K_2 \oplus K_3$ gives $T_1(S) = \ker(\operatorname{colim} L \to \mathbb{Z}^3) = \ker(\operatorname{colim} L \xrightarrow{\varepsilon} \mathbb{Z})$, because the map is $l \mapsto (\varepsilon l, -\varepsilon l, \varepsilon l)$; it also gives $T_i(S) = T_{i-1}(L)$ for $i \geq 2$. Acyclicity of K then gives the asserted formulas. They are the same positive derived-group descriptions as for Q_a in Theorem B.4, without asserting $C \cong Q_a$.

The identity $\kappa = \psi(\rho_1, -\rho_2, \rho_3)$ has augmentation vector $(1, -1, 1)$, which is the connecting image of $1 \bmod \varepsilon(L)$ by the preceding proof. Common successors make absence of an affine equalizer equivalent to $\varepsilon(L) \neq \mathbb{Z}$. The bar complex computes $T_1(C)$, so the nonzero class is represented by a bar 1-cycle. Lift its finitely many coefficients through the pointwise epimorphism $K \to C$; its bar boundary then lies in S and represents $[\kappa]$. This is the precise coefficient interpretation of the last sentence of Lead 5. An ordinary nonzero class in $H_1(BA; \mathbb{Z})$ cannot be inferred: totality gives BA weakly contractible. □

C The Adámek–Koubek–Reiterman construction and finite tests

C.1 Finite tests and comparison of links

Lemma C.1 (Proved: finite-poset representation of spherical maps)**.** *Let E be a small category, let $n \geq 1$, and let $f : S^n \to |NE|$ be continuous. There is a finite triangulated n-sphere K, an iterated barycentric subdivision K', and a functor $v : D \to E$, where D is the face poset of K', such that Bv represents the homotopy class of f under the subdivision homeomorphism $BD \cong S^n$.*

Proof. We use the standard subdivision–extension adjunction $\mathrm{Sd} \dashv \mathrm{Ex}$. The natural map $NE \to \mathrm{Ex}^\infty NE$ is a weak equivalence and the target is Kan. Choose the boundary of an ordered $(n+1)$-simplex as the finite simplicial sphere. In a Kan complex every based nth homotopy class is represented on this boundary by placing an n-simplex with all faces at the basepoint on one face and placing degenerate basepoint simplices on the other faces. Thus we can represent the given class on an actual finite triangulated sphere, whose iterated subdivisions remain triangulated spheres. Since its set of nondegenerate simplices is finite, the representative factors through $\mathrm{Ex}^r NE$ for a finite r. Its adjoint is a simplicial map from an iterated subdivision of the chosen sphere to NE. Subdividing once more gives the nerve of a face poset, and composition with the last-vertex map gives a simplicial map $ND \to NE$. The nerve functor is fully faithful, so this map is Nv for a unique functor v. The realization of the last-vertex map is homotopic to the subdivision homeomorphism; the usual subdivision–extension comparison consequently represents the original homotopy class. This is an approximation of the map, not an identification of E with a poset. □

Proposition C.2 (Proved: Lead 6)**.** *Let A be a small nonempty connected category. For a functor $u : D \to A$, define the covariant A-set*

$$H_u(a) = \pi_0(u \downarrow a)$$

and the functor $j_u : D \to \mathrm{El}_A H_u$ by $j_u(d) = (u(d), [d, \mathrm{id}_{u(d)}])$. Then A is total if and only if Bj_u is nullhomotopic on every component for every finite poset D and every such u. It suffices to use face posets of finite triangulated spheres of positive dimension.

Proof. For precision, an object of $u \downarrow a$ is $(d, f : u(d) \to a)$, and an arrow $(d, f) \to (d', f')$ is $\delta : d \to d'$ with $f = f'u(\delta)$. Postcomposition defines the A-action on H_u, and $j_u(\delta) = u(\delta)$.

Let $H : A \to \mathrm{Set}$ and $v : D \to \mathrm{El}_A H$ be arbitrary. Write $v(d) = (u(d), h_d)$. Define a natural transformation of A-sets $\lambda : H_u \to H$ by

$$\lambda_a[d, f] = H(f)h_d.$$

The equation $H(u(\delta))h_d = h_{d'}$ makes this independent of representatives in $\pi_0(u \downarrow a)$; postcomposition proves naturality. The induced functor $\mathrm{El}_A \lambda : \mathrm{El}_A H_u \to \mathrm{El}_A H$ satisfies the literal equality

$$v = (\mathrm{El}_A \lambda) j_u$$

on objects and arrows.

If A is total, each component of $\mathrm{El}_A H_u$ has weakly contractible CW realization and hence contractible realization by Whitehead's theorem. Therefore each component restriction of Bj_u is nullhomotopic.

Conversely suppose the asserted nullhomotopy condition holds for sphere face posets. Given H, a component E of $\mathrm{El}_A H$, and a map $S^n \to BE$ with $n \geq 1$, Lemma C.1 represents its homotopy class by a functor $v : D \to E$. Compose with $E \hookrightarrow \mathrm{El}_A H \to A$ to obtain u. The displayed factorization and the hypothesis make Bv nullhomotopic. All homotopy groups of BE therefore vanish. Since E is nonempty and connected, BE is weakly contractible. This proves totality. The empty poset adds no condition, and the separate components of S^0 add only the automatic pointwise nullhomotopies. □

Proposition C.3 (Proved: Lead 7)**.** *Let A be small, $R : A^{\mathrm{op}} \to \mathrm{Set}$ a presheaf, and A' the category obtained by adjoining one object Z with $A'(x, Z) = R(x)$, $A'(Z, Z) = \{\mathrm{id}_Z\}$, and no arrows from Z to old objects. Let $H' : A' \to \mathrm{Set}$ and $H = H'|_A$. The old comma link L_h over (Z, h), for $h \in H'(Z)$, is a union of components of the two-sided category of elements* $\mathrm{El}(R, H)$.

Suppose a functor $\widetilde{u} : \widetilde{D} \to A$ and an isomorphism

$$R \cong \operatorname*{colim}_{d \in \widetilde{D}} A(-, \widetilde{u}(d))$$

are specified, and every component of every $x \downarrow \widetilde{u}$ is weakly contractible. Then the canonical functor

$$F : \mathrm{El}_{\widetilde{D}}(\widetilde{u}^* H) \longrightarrow \mathrm{El}(R, H)$$

is a weak equivalence, as is its restriction over each h. If $n = \dim N\widetilde{D} < \infty$, then $H_j(L_h; \mathbb{Z}) = 0$ for $j > n$, and

$$H_j(\mathrm{El}_A H; \mathbb{Z}) \longrightarrow H_j(\mathrm{El}_{A'} H'; \mathbb{Z})$$

is injective for $j > n$.

Proof. An object of $\mathrm{El}(R, H)$ is (x, r, k) with $r \in R(x)$ and $k \in H(x)$. An arrow to (y, s, l) is $f : x \to y$ satisfying $R(f)s = r$ and $H(f)k = l$. The function

$$(x, r, k) \longmapsto H'(r)k \in H'(Z)$$

is constant along every arrow. Its inverse image of h is literally L_h; it is consequently a union of components. No injectivity condition on the maps of H' is used.

The specified presheaf presentation identifies $R(x) = \pi_0(x \downarrow \widetilde{u})$. Define

$$F(d, k) = (\widetilde{u}(d), [d, \mathrm{id}], k), \qquad F(\delta) = \widetilde{u}(\delta).$$

For $e = (x, r, k)$, an object of $e \downarrow F$ is exactly a pair $(d, f : x \to \widetilde{u}(d))$ representing r: the source H-element in F is forced to be $H(f)k$. The arrow conditions are exactly those of $x \downarrow \widetilde{u}$. Thus $e \downarrow F$ is isomorphic to the component indexed by r. The assumed weak contractibility and Quillen's Theorem A give the weak equivalence of the specified functor F. Restricting to the union of source components that map to L_h leaves these comma categories unchanged, so the restriction is a weak equivalence too.

A nonidentity arrow in a category of elements cannot lie over an identity. Every nondegenerate simplex of $N\,\mathrm{El}_{\widetilde{D}}(\widetilde{u}^{*}H)$ therefore projects to a nondegenerate simplex of $N\widetilde{D}$. Its dimension is at most n, and the same holds for each union of its components. This proves the asserted vanishing of link homology.

Every nondegenerate simplex of $N\,\mathrm{El}_{A'}\,H'$ is either old or ends at exactly one new object (Z,h). In the latter case its preceding chain, including its last arrow into (Z,h), is a simplex of NL_h. Thus the relative chain complex is the direct sum over h of the augmented link complexes shifted by one. The long exact sequence contains

$$\bigoplus_{h} H_j(L_h;\mathbb{Z}) \longrightarrow H_j(\mathrm{El}_A\,H;\mathbb{Z}) \longrightarrow H_j(\mathrm{El}_{A'}\,H';\mathbb{Z}),$$

which proves injectivity for $j > n$. Infinite sets of new elements cause no problem because chains are finite sums and homology commutes with direct sums. □

Remark C.4 (Proved: the limitation of the one-dimensional test)**.** Under Proposition C.3's hypothesis, a necklace diagram of dimension one or a sum diagram of dimension zero cannot kill an old H_2 class. The comma hypothesis is essential. In a quasi-filter necklace for $f,g:T\to S$, if $fc=gc$ for an arrow $c:x\to T$, the copies of c and fc form a circular 1-cycle in $x\downarrow\widetilde{u}$. This comma category has dimension at most one, so that nonzero cycle cannot be a boundary. Its component is not weakly contractible. In the actual-path repair, the explicit identity $\alpha_{4k}v_1=\alpha_0v_1$ supplies this failure for a subsequent quasi-filter of (α_0,α_{4k}) at $x=V$. Consequently this proposition does not, by itself, prove persistence through all stages of the repaired construction.

C.2 The interval-type candidate and necessary conditions

Proposition C.5 (Proved: Lead 8)**.** *Let* Ty *be Grothendieck's category of finite interval types with endpoint-preserving functors that are nondecreasing on the numerical vertex order, as in VII manuscript pages 31–32 and PRETOT Definition E.1. Its opposite is the category of finite words in* $+,-$ *and weakly increasing sign-preserving maps of letters. In* $\mathsf{Ty}^{\mathrm{op}}$ *the two maps*

$$e=(1,3),\quad c=(1,1):(++)\longrightarrow(+-+)$$

have no completion to a commutative square. They have no affine equalizer over any nonzero commutative unital ring. Therefore $\mathsf{Ty}^{\mathrm{op}}$ *is neither total nor exact, and its colimits fail to be exact over every such ring.*

Proof. We first verify the variance and the complete combinatorial identification. A Ty-map $\tau\to\rho$ has a nondecreasing vertex function $\phi:\{0,\dots,|\tau|\}\to\{0,\dots,|\rho|\}$ preserving both endpoints. For every letter j of ρ there is a unique letter i of τ such that $\phi(i-1)<j\le\phi(i)$. Send j to this i. The resulting map of letters is weakly increasing. It preserves signs because the nonempty path traversed by a source edge must have the same sign throughout. Conversely, from a weakly increasing sign-preserving map f from the letters of ρ to those of τ, define

$$\phi(i)=|\{j:f(j)\le i\}|.$$

These operations are inverse, also when either word is empty. The construction commutes with composition: the source edge traversing a target edge is found by successive pullback of edge indices. Thus we have an isomorphism of categories on the fixed word objects, with the stated opposite variance.

Set $p=(++)$ and $q=(+-+)$. For every word σ and arrow $g:q\to\sigma$, monotonicity and the different signs of adjacent letters imply $g(1)<g(2)<g(3)$. Thus $ge:p\to\sigma$ is injective,

whereas hc is constant for every $h : q \to \sigma$. Equality $ge = hc$ is impossible, proving failure of condition I.

Let $R \neq 0$ be a commutative unital ring. Define an R-linear functional

$$\ell_\sigma : R[\mathsf{Ty}^{\mathrm{op}}(p, \sigma)] \to R$$

by assigning value 1 to an injective two-letter map and value 0 to a constant two-letter map. These exhaust the possibilities. For every finite linear combination $t \in R[\mathsf{Ty}^{\mathrm{op}}(q, \sigma)]$ we have

$$\ell_\sigma(te) = \mathrm{aug}(t), \qquad \ell_\sigma(tc) = 0.$$

The equation $te = tc$ therefore implies $\mathrm{aug}(t) = 0$, which excludes augmentation one. Totality implies integral condition IV by the pair quotient calculation verified in TOTEX A.4, and exactness over R implies the affine criterion of Isbell and Mitchell (1976, Theorem 3.1 and the remark on p. 296). Both conclusions follow. The empty word is initial in $\mathsf{Ty}^{\mathrm{op}}$, so the category is nonempty and connected.

We checked the adopted definition against PRETOT E.1 and the supplied manuscript transcription. At VII ms. 31 the added restriction is explicitly that the functors are "croissants sur les objets". Our independent program `code/topology_ty_verify.py` compares all 961 pairs of sign words of length at most four, finding no discrepancy between vertex maps and letter maps; checks 2731 compositions through length three; and tests 7937 arrows from $(+ - +)$ to all 1023 words of length at most nine. None gives a square completion or overlapping constant/injective supports. These finite checks supplement the arbitrary-word and arbitrary-ring proof above. □

Proposition C.6 (Proved: Lead 9)**.** *Let A be a separator, and let $a_0, \dots, a_k : p \to q$ have no integral affine equalizer, where k is minimal among all such families in A. Then $k \geq 2$ and the following hold:*

(a) $A(q, p) = \varnothing$;

(b) the cardinalities $|A(p, x)|$ have no finite uniform bound;

(c) the full cosieve C on objects receiving an arrow from p is again a separator, containing this family, and p is weakly initial in C;

(d) for every arrow $g : q \to x$, the arrows $ga_0, \dots, ga_k$ are pairwise distinct.

Proof. Totality supplies I and IV; thus pairs have affine equalizers and $k \geq 2$.

For (a), if $q \to p$ existed, Isbell and Mitchell (1976, Corollary 4.3, p. 297) would equalize the finite family by an ordinary arrow. We opened the corollary and its preceding Theorem 4.1 and Corollary 4.2: the second assertion applies exactly with source p and target q, under $A(q, p) \neq \varnothing$, and requires no weakly terminal object. The ordinary equalizer would also be affine, a contradiction.

For (b), the representable A-set $A(p, -)$ is indecomposable because its element category p/A has initial object id_p. A finite bound on its values would allow Isbell and Mitchell (1976, Proposition 4.5, pp. 297–298) to apply under I and V; IV and V are equivalent by their necklace criterion. It gives condition II for this representable. Inductively equalizing pairs then ordinarily equalizes every finite family in a value, again a contradiction. The source proof was opened: eventual equality defines a congruence using I; a quotient value of maximal finite cardinality, together with the necklace identity, forces all quotient elements equal. The bound is needed only for this one representable, not simultaneously for all of them.

For (c), C contains p, q and is a cosieve. Every C-set H extends to an A-set by assigning the empty set outside C: there are no arrows from C to its complement, and arrows entering

C receive the unique empty-source function. Its category of elements is literally $\mathrm{El}_C H$, so totality of A proves totality of C. The object p maps to every object of C, hence C is nonempty and connected. Every affine arrow in C is also one in A, so the displayed family remains unequalizable. Thus C is a separator.

For (d), suppose $ga_i = ga_j$ for $i \neq j$. The set of distinct arrows among the ga_l has at most k members. By the global minimality assumption, it has an affine equalizer $t : x \to y$. The composite $tg : q \to y$ is affine, because $\mathrm{aug}(tg) = \mathrm{aug}(t)\,\mathrm{aug}(g) = 1$, and it equalizes all a_l. This contradiction proves pairwise distinctness. Global minimality among families in A is used here; minimality only among subfamilies of the chosen family would not suffice. □

C.3 The checked expansion invariant and a sharper obstruction

Proposition C.7 (Known, with source corrections: the repaired AKR invariant)**.** *Let $K^* = \bigcup_m K_m$ be the fresh-target construction of Adámek et al. (1981, Construction C.1), with the actual-path repair in TOTEX Proposition C.1, and any admissible sequence $a_0 = 2 > a_1 > \cdots > 3/2$ and quasi-filter parameters. For an object $X \in K_m$, an arrow $g : Y \to X$, and a finite family of pairwise distinct arrows $h_i : W \to X$, we have:*

(a) if $|\operatorname{im} h \cap \operatorname{im} g| \geq 2$ for an arrow $h : W \to X$, then $h = gh'$ for an arrow $h' : W \to Y$;

(b) if $|\operatorname{im} h_i \cap \operatorname{im} g| \leq 1$ for every i, then $|\bigcup_i \operatorname{im} h_i \setminus \operatorname{im} g| \geq a_m r$;

(c) for every nonempty such family, $|\bigcup_i \operatorname{im} h_i| > 3r/2$.

The maps in (a) and the equality $h = gh'$ are maps of the concrete category of finite sets and injections, not merely factorizations of sets after forgetting the category.

Verification of the cited induction. We opened the full proof of Theorem C.2 on pp. 385–389, including the induction on $(c_m), (d_m)$ that was not completed in the previous review. These are precisely (a),(b). We record the operative steps so that the use of the theorem is explicit.

At the initial stage the only nonidentity arrows are the three points of W, so (a),(b) hold with $a_0 = 2$. For a new quasi-filter target Z, every old arrow into Z factors through one of its copies $\alpha_i : T \to Z$. If a W-map meets a specified copy in at least two points, overlap condition (10) allows us to move it one neighboring copy toward that copy. Condition (11) identifies the neighboring overlap with the image of the relevant map $S \to T$. Applying the old (a) to that overlap gives a morphism $W \to S$, and hence the required transported W-map in the neighboring copy. Repetition moves the whole arrow into the specified copy. The old (a) there proves the new (a), including the categorical factorization.

For (b), assign each selected W-map to its first possible copy and partition the cyclic list of s copies into blocks of $3n$. Remove a block carrying the fewest assigned arrows; at most $r\,3n/s$ arrows are lost. Traverse the remaining copies from the two ends toward the removed block. At each step, (10) puts all intersection with the previously visited copies inside the neighboring overlap. A newly assigned W-map cannot meet that overlap in two points: (a), (11), and monicity would make it factor through the preceding copy, contrary to its assignment. The old (b) therefore contributes at least a_m new points for every retained arrow. Hence

$$\left|\bigcup_i \operatorname{im} h_i \setminus \operatorname{im} g\right| \geq a_m r \left(1 - \frac{3n}{s}\right) \geq a_{m+1} r,$$

where the last inequality is exactly the parameter restriction in Construction C.1. The two traversal ranges and their indices were checked directly on the page images on pp. 387–389. For a new disjoint sum, arrows factor through one summand, and the two summands contribute disjoint sets; the old bounds give the claimed bound immediately.

For (c), distinct W-maps meet in at most one point: otherwise (a) factors one through the other, and the only endomorphism of W is its identity. Applying (b) with $g = h_1$ to the remaining $r - 1$ arrows yields

$$\left|\bigcup_{i=1}^{r} \operatorname{im} h_i\right| \geq 3 + a_m(r-1) > \frac{3}{2}r.$$

This also justifies the union bound used in the disjoint-sum case of the printed induction.

The actual-path repair supplies (9)–(11); we checked the formula and the interval overlap verification in TOTEX Proposition C.1. Two further index misprints in the source do not affect the induction: on p. 387, the definition of L must use $\{1, \dots, r\}$, not $\{1, \dots, n\}$; r counts selected W-maps and n is the unrelated path bound. On p. 388, "Then (g_{m+1}) follows" must refer to (g_t). These corrections were confirmed against the page images, not inferred only from OCR. □

Proposition C.8 (Proved: pointwise splitting of the AKR column)**.** *Let R be a nonzero commutative unital ring, let $P_x^R = R[K^*(x, -)]$, and define a morphism of covariant R-module diagrams*

$$d^T : P_W^R \longrightarrow (P_V^R)^2, \qquad t \longmapsto \big(t(v_2 - v_1), t(v_3 - v_1)\big).$$

For every object X, d_X^T is a split injection of finite free R-modules, and its cokernel $Q(X)$ is finite free. No natural splitting is asserted. For the diagram $Q = \operatorname{coker} d^T$, derived colimits satisfy

$$\operatorname*{colim}_{K^*} Q \cong R^2, \qquad L_1 \operatorname*{colim}_{K^*} Q \cong R, \qquad L_i \operatorname*{colim}_{K^*} Q = 0 \quad (i \geq 2).$$

Thus the failure of exactness is witnessed by an objectwise free diagram with a projective resolution of length one.

Proof. Every object X is introduced at a finite stage. No later stage adds arrows between old objects, so $K^*(W, X)$ and $K^*(V, X)$ are finite. For a nonempty set S of r columns, Proposition C.7(c) says the corresponding $3r$ incidences gv_i use more than $3r/2$ points. Hence some point belongs to exactly one of those incidences. If its color is v_2 or v_3, the corresponding row in the first or second coordinate has a sole entry $+1$ among these columns. If its color is v_1, either corresponding row has a sole entry -1.

Choose such a row, remove its unique column, and repeat. The selected rows are distinct, because each previously selected row is zero on all remaining columns. Ordering the rows and columns in this elimination order gives a triangular square minor with diagonal entries ± 1. It is invertible over $\mathbb{Z}$, hence over every R. Projection onto the selected rows followed by its inverse is a left inverse of d_X^T. Eliminating these selected coordinates identifies the quotient with a free module on the unselected rows. This proves pointwise splitting and freeness without selecting splittings natural in X.

Thus

$$0 \longrightarrow P_W^R \xrightarrow{d^T} (P_V^R)^2 \longrightarrow Q \longrightarrow 0$$

is an exact sequence of diagram modules, and its two left terms are projective. For every object x, the augmentation identifies $\operatorname{colim} P_x^R$ with R: the category x/K^* has initial object id_x. Under these identifications the induced map $R \to R^2$ is zero, since both coordinate differences have augmentation zero. Applying colimit to this length-one projective resolution gives exactly the three displayed derived-colimit groups. This is compatible with non-totality, but it does not prove non-totality: the coefficient diagram Q has not been shown to lie in the acyclic class generated by permutation diagrams. □

C.4 Status of the all-stage homology assertion

Unresolved statement C.9 (Part C.1: persistence through every admissible stage)**.** For every admissible repaired AKR construction, is

$$H_2\big(\mathrm{El}_{K^*}(K^*(V,-)/(cv_1 \sim cv_2)); \mathbb{Z}\big) \neq 0?$$

The universal assertion is not established in this interrupted record. The initial repaired first-stage class and its injection through the entire second stage were checked in the supplied TOTEX proof; they do not by themselves imply persistence in the union. Proposition C.3 does not apply to all future quasi-filters, by Remark C.4. No separator is claimed.

Proposition C.10 (Proved by finite enumeration: six additional component tests)**.** *Starting from the repaired first stage with $s = 45$, take the quasi-filter targets for (v_1, v_2) and (v_1, v_3). At each of these targets order its 45 maps from W by their concrete tuples, and choose the index pairs $(0,1), (0,44), (0,22)$. For each of these six cases add only the selected second-stage quasi-filter, with $s = 153$. For the fixed pair quotient H, every generating map α_i induces an injection on H-values. For each target, all 11628 unordered pairs of its 153 generating maps have acyclic induced pair graphs on H-values; no induced path is longer than its underlying point-set counterpart. Both maximum path lengths, measured in vertices, are three. There are no reversed two-element $H(W)$-actions among the 6732 incoming W-maps at any of the six new targets.*

Proof. The program `code/topology_akr_component_probe.py` uses the independently supplied stage implementation, retained as `code/topology_akr_reference.py`. It builds all concrete maps of each specified substage and verifies the neighbor identities and repaired overlap property (11). For every object, a union–find quotient identifies gv_1 with gv_2 for every incoming W-map; this is exactly H. The induced maps on quotient classes are computed on every point and checked to be well-defined. Injectivity is tested by comparing image cardinalities. For every ordered pair selected for comparison, removing fixed arrows leaves the partial-injection graph; traversal from all indegree-zero vertices detects path lengths, and any remaining vertices would detect directed cycles. Finally a hash set of the induced maps $H(W) \to H(Z)$ is tested against its reversed tuples. The output records all six complete checks. This proposition is deliberately confined to those six finite targets. It is neither a computation of the full second stage nor an induction through later stages. □

The remaining prospective induction is stronger than these tests: one would need to control the homology maps of every future link, or show that the pair quotient preserves injectivity and adequate bounds for every relevant morphism chain. A path in the quotient can join components through different copies of an old object; the pointwise injectivity of the underlying set maps alone does not exclude that event. The common-successor or down-set argument also stops when a sum contains both initial disk targets, because that down-set already has the nonzero second-homology class.

C.5 The homogeneous Ramsey candidates

Definition C.11 (Known: finite-family Ramsey convention)**.** We say that a small category A has the finite-family Ramsey property if, for every finite family $a_0, \dots, a_k : p \to q$ and every integer $N \geq 2$, there is an object r such that every map $\chi : A(p,r) \to \{1, \dots, N\}$ admits an arrow $b : q \to r$ with $\chi(ba_0) = \cdots = \chi(ba_k)$. The usual categorical Ramsey property makes all of $bA(p,q)$ monochromatic and therefore implies this condition.

Theorem C.12 (Proved: a Ramsey sufficient condition for integral exactness)**.** *Let A be a small category in which every finite family of objects has a common successor. If A has*

the finite-family Ramsey property, then A satisfies condition III over $\mathbb{Z}$. If A also satisfies condition I, then $\operatorname{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ *is exact.*

Proof. Fix $a_0, \dots, a_k : p \to q$. For $k = 0$ the identity of q suffices, so we assume $k \geq 1$. For every object r we define a homomorphism of abelian groups

$$D_r : \mathbb{Z}[A(q,r)] \longrightarrow \mathbb{Z}[A(p,r)]^k, \qquad D_r(t) = \big(t(a_i - a_0)\big)_{i=1}^k,$$

and put $E = \sum_r \operatorname{aug}(\ker D_r) \subseteq \mathbb{Z}$. We first verify that $E = \mathbb{Z}$ if and only if this family has an integral affine equalizer. One direction is immediate. For the other, write $1 = \sum_{j=1}^m \operatorname{aug}(t_j)$ with $t_j \in \ker D_{r_j}$, absorbing any integer multipliers into the t_j. Choose arrows $u_j : r_j \to s$ to a common successor. Then $t = \sum_j u_j t_j \in \ker D_s$ and $\operatorname{aug}(t) = 1$.

Suppose that there is no affine equalizer. We have $E = d\mathbb{Z}$ with either $d = 0$ or $d \geq 2$. Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with its translation-invariant circle metric. Choose $c = 1/2 \pmod{\mathbb{Z}}$ when $d = 0$, and $c = 1/d \pmod{\mathbb{Z}}$ when $d \geq 2$. Thus $c \neq 0$ and $ec = 0$ for every $e \in E$. For every r the formula

$$\lambda_r(D_r t) = \operatorname{aug}(t)c$$

defines a homomorphism $\lambda_r : \operatorname{im} D_r \to \mathbb{T}$: any two preimages differ by an element of $\ker D_r$, whose augmentation kills c.

We extend λ_r to a homomorphism $\widetilde{\lambda}_r : \mathbb{Z}[A(p,r)]^k \to \mathbb{T}$. Here we use divisibility of the circle group. For completeness, a homomorphism from a subgroup B' of an abelian group B to a divisible group extends across one additional element $b \in B$: if no positive multiple of b lies in B', choose its image arbitrarily; otherwise take the least positive n with $nb \in B'$ and choose an image x with nx equal to the given image of nb. The extension to $B' + \mathbb{Z}b$ is then well-defined. Zorn's lemma gives an extension to B.

Let $\phi_i^r : A(p,r) \to \mathbb{T}$ be the function obtained by restricting $\widetilde{\lambda}_r$ to basis elements in its ith summand. For every $b : q \to r$ we have

$$\sum_{i=1}^{k} \big(\phi_i^r(ba_i) - \phi_i^r(ba_0)\big) = c. \tag{1}$$

Put $\eta = \operatorname{dist}_{\mathbb{T}}(c, 0) > 0$ and choose $\delta > 0$ with $k\delta < \eta$. Partition $\mathbb{T}^k$ into finitely many cells such that two points in one cell have distance less than δ in every coordinate; we may arrange that the number N of cells is at least two. Choose r from the finite-family Ramsey property for this N. Color $h : p \to r$ by the cell containing $(\phi_1^r(h), \dots, \phi_k^r(h))$. There is $b : q \to r$ for which $ba_0, \dots, ba_k$ have one color. Every summand on the left of (1) has circle distance less than δ from zero. Their sum has distance less than $k\delta < \eta$, contradicting (1). Therefore $E = \mathbb{Z}$, proving III. The last assertion follows from Isbell and Mitchell (1976, Proposition 3.3). □

Corollary C.13 (Proved: the Ramsey candidates in route C4 are exact)**.** *The categories of nonempty finite linearly ordered simple graphs and of nonempty finite linearly ordered triangle-free graphs, with order-preserving induced embeddings, are exact over $\mathbb{Z}$ and hence over every commutative ring. They are total and are not filtered. More generally, every nonempty rigid Ramsey amalgamation class of finite structures with the joint embedding property has exact abelian colimits.*

Proof. We take a small skeleton in each case. Joint embedding gives common successors, and amalgamation is condition I. The ordinary Ramsey property for ordered graphs is Hubička and Nešetřil (2019, Corollary 4.3); for ordered triangle-free graphs it is Hubička (2025, Theorem 15). These statements concern all finite colorings and order-preserving induced embeddings, exactly the morphisms used here. Because the structures are rigid, copies and embeddings have no automorphism ambiguity. Theorem C.12 applies.

We verify the amalgamation hypothesis explicitly. For a span of finite ordered graphs, take the union over the common induced graph, put no new edges between the two new parts, and extend the two agreeing linear orders to a linear order of the union. This is an induced order-preserving amalgam. Every triangle in this union lies in one of the original graphs, so the construction also works in the triangle-free class. Disjoint ordered union gives joint embedding. Both categories fail ordinary filteredness: the two embeddings of a one-vertex graph into a fixed two-vertex graph remain distinct after every injective map. Integral affine equalizers give affine equalizers after every change of commutative coefficients. Totality follows from the exactness implication in R2. The same argument proves the stated general consequence for rigid Ramsey amalgamation classes. □

Remark C.14 (Unresolved: scope of the Ramsey argument)**.** We have not proved that totality implies the finite-family Ramsey property. Thus Theorem C.12 closes the named route C4 candidates, but does not settle the main implication for arbitrary total categories. We make no priority claim: searches for "Ramsey exact colimits", "Ramsey affinization", "Ramsey Isbell Mitchell", and related phrases did not identify this implication, and Isbell's 1974 full text was not obtained.

D Class inclusions, manuscript corrections, and sources

D.1 The class diagram and the manuscript corrections

Proposition D.1 (Known inclusions; two adjacent comparisons unresolved)**.** *On nonempty connected small categories, the verified conclusions are*

$$\{\textit{filtered}\} \subsetneq \{\textit{exact}\} \subseteq \{\textit{total}\} \subseteq \{\textit{quasi-filtered}\}.$$

The exact and quasi-filtered classes differ. This record does not decide either exact = total or total = quasi-filtered separately.

Proof. We use the checked implications R1–R4 in the supplied prompt. The category Δ_{inj} of nonempty finite linearly ordered sets and increasing injections is exact by Isbell and Mitchell (1973, Theorem 3), and is not filtered: its two maps from a singleton to a two-element chain cannot be equalized by an injection. Exactness implies totality by R2. The quasi-filtered, nonexact category of Adámek et al. (1981, Construction C.1 and Theorem C.2) separates the outer two classes. Part C records the checked proof of that theorem and the repairs used for its parameters. Its totality has not been proved or refuted in this return. Thus its nonexactness does not identify which adjacent inclusion is strict.

We also read the full interpolation verification in TOTASPH C.8–C.10. The integral coordinate change $x_i = z_i - z_{i+1}$ reduces compatible coface data to coordinate-hyperplane interpolation, whose alternating sum preserves homogeneity and augmentation one. The prescriptions $D_i C_{n,k} = C_{n-1,2k} \prod_{l \neq i} y_l^k$ agree on adjacent intersections directly and on nonadjacent intersections by the inductive normalized-face identity. Gap encoding turns the resulting polynomial into an affine arrow equalizing all cofaces. The simplicial identities then equalize every ordinary parallel family. This confirms the exactness used here without treating the unopened 1974 body as a proof.

We re-read the needed proof steps of R1, R2 and R4 in TOTASPH C.3–C.6 and C.11 and TOTEX A.1–A.4. In the sufficiency direction of R1 used below, for an ordinary final $f : A \to B$ the comma category $b \downarrow f$ is exactly the category of elements of $a \mapsto B(b, f(a))$. Thus the action condition makes each comma category weakly contractible, and Theorem A applies to the specified map Bf. For the implication exact $\Rightarrow$ total in R2, the functor

$$\Sigma_H : \mathrm{Ab}^{\mathrm{El}_A H} \longrightarrow \mathrm{Ab}^A, \qquad (\Sigma_H F)(a) = \bigoplus_{h \in H(a)} F(a, h),$$

is exact and has a canonical colimit comparison $\operatorname{colim}_{\mathrm{El}_A H} F \cong \operatorname{colim}_A \Sigma_H F$. Hence all action components have exact colimits, and their constant coefficient bar complexes are integrally acyclic. The inherited-action construction in the next proof lets us apply the same conclusion to every connected covering of every such component. In particular the cover belonging to a cyclic subgroup K of its fundamental group has $H_1 = K_{\mathrm{ab}} = K = 0$. Every cyclic subgroup is trivial; the component is simply connected and acyclic, and Hurewicz gives weak contractibility. This verifies the crucial integral-to-homotopy step without presupposing the converse to R2.

For R4, if two objects have no common successor, their disjoint principal cosieves U and V give the action formed by identifying the two constant values precisely on $U \amalg V$. Its nerve is two copies of NA glued on $N(U \amalg V)$, because arrows cannot leave a cosieve. Each cosieve is connected. Mayer–Vietoris gives $H_1 = \mathbb{Z}$, contradicting totality. Applying the same argument to $p/A = \mathrm{El}_A A(p, -)$ completes every span out of p to a square. For $u, v : p \to q$ the quotient action $H = A(p, -)/(cu \sim cv)$ has free presentation $P_q \xrightarrow{v-u} P_p \to \mathbb{Z}[H] \to 0$. If $K = \ker(P_q \to P_p)$, the two short exact sequences obtained by inserting its image give $H_1(B\,\mathrm{El}_A H; \mathbb{Z}) = \mathbb{Z}/\operatorname{aug}(\operatorname{colim}_A K)$. Vanishing gives an augmentation-one finite sum in K; common successors move its finitely many supports to a single target. This is IV, together with I proved above. In particular no step replaces a kernel in a direct sum by an intersection of kernels. □

Proposition D.2 (Refuted conjecture; known R6 repairs of II ms. 36–36ter)**.** *The conjecture printed at II ms. 36ter,*

$$X \text{ totalement } W_\infty\text{-asphérique} \Longrightarrow X \text{ filtrante},$$

is false. For nonempty connected small categories its following strengthenings are true:

(i) a total sifted category is filtered;

(ii) for any fixed integer $n \geq 1$, a total category A is filtered if $\operatorname{hocolim}_A K(M(a), n)$ *is n-truncated for every covariant diagram $M : A \to \mathrm{Ab}$.*

Filtered categories satisfy both additional conditions. These are the checked R6 results, not a resolution of totality versus exactness.

Reading and proof check. We read II ms. 36, 36bis and 36ter in the supplied Künzer–Malgoire–Maltsiniotis transcription (Grothendieck, 2001). The category Δ_{inj} just recalled refutes the conjecture. This uses the published example and makes no claim of a new counterexample. The original body of Isbell's 1974 paper was not obtained in this session; its proof is therefore not represented as having been read.

There are two distinct assertions on ms. 36. The proposed Proposition 2.19, which infers totality from finality of the diagonal, is expressly footnoted "Faux." The subsequent question asks whether totality implies that final-diagonal condition, or at least PS 1′. With the nonemptiness convention, finality of the diagonal is siftedness. Totality does not imply siftedness, since (i) and Δ_{inj} would otherwise imply filteredness. It does imply PS 1′: any two objects have a common successor. This follows already from the span-completion condition I in R4. The printed corollary claiming totality of Δ^{op} is false as well. For example, the simplicial circle, regarded as a functor $\Delta^{\mathrm{op}} \to \mathrm{Set}$, has a category of elements whose classifying space is weakly equivalent to the circle, by the usual simplex-category realization comparison. Its first homology is $\mathbb{Z}$, so it violates the action criterion R1.

For clarity we spell out the short proof of (i), also proved in the supplied PRETOT Lemma F.3 and rechecked in TOTEX D.4. Totality passes to every connected component C of $\mathrm{El}_A S$: for $T : C \to \mathrm{Set}$ the functor

$$R(a) = \coprod_{\substack{s \in S(a) \\ (a,s) \in C}} T(a, s) \quad (a \in A)$$

has structure maps induced by those of S and T, and there is an actual isomorphism $\mathrm{El}_A R \cong \mathrm{El}_C T$. In particular such a nonempty connected C has common successors, by R4. If A is sifted, its cospan category

$$C_{p,q} = \mathrm{El}_A(A(p,-) \times A(q,-))$$

is nonempty and connected for every p, q. This is the criterion of Gabriel and Ulmer (1971, Example 15.2(c), proof in §15.9); both the statement and that proof were opened in this session. Given $u, v : p \to q$, a common successor in $C_{p,q}$ of (q, u, id_q) and (q, v, id_q) supplies maps $f, g : q \to r$ with $fu = gv$ and $f\,\mathrm{id}_q = g\,\mathrm{id}_q$. Hence $f = g$ and $fu = fv$. Siftedness itself supplies common successors in A, so A is filtered. Every map in this argument is an ordinary arrow of A.

We also re-read the complete abelian and fixed-n proofs of R6 in the supplied TOTEX D.5 and D.7. Their relevant comparisons are as follows. Write $G = \mathrm{colim}_A^{\mathrm{Ab}} M$. If $n \geq 2$, totality gives $BA \simeq *$; connectivity of the bar construction and its degree-n edge map show that

$$X_M = \operatorname*{hocolim}_A K(M(a), n) \longrightarrow K(G, n)$$

is a weak equivalence under the assumed n-truncation. The reduced bar spectral sequence has row n equal to M, no lower rows and zero row $n+1$, since $H_{n+1}(K(L,n); \mathbb{Z}) = 0$ for every abelian group L. Thus

$$L_1 \operatorname*{colim}_A M = H_{n+1}(X_M; \mathbb{Z}) = 0.$$

This proves exactness, rather than assuming the unresolved converse to R2. Exactness then collapses the same homology spectral sequence in every degree. Apply the resulting canonical homology comparison to $M \oplus N$ and take the cross summand in degree $2n$ of $K(M,n) \times K(N,n)$. The integral Künneth formula gives

$$\operatorname*{colim}_A (M \otimes_{\mathbb{Z}} N) \xrightarrow{\cong} (\operatorname*{colim}_A M) \otimes_{\mathbb{Z}} (\operatorname*{colim}_A N).$$

No Tor summand occurs in that cross degree, since both positive indices are at least n. Taking $M = \mathbb{Z}[A(p,-)]$ and $N = \mathbb{Z}[A(q,-)]$ makes this the free abelian map from $\pi_0 C_{p,q}$ to a singleton. Consequently $C_{p,q}$ is nonempty and connected. Hence A is sifted, and (i) applies.

For $n = 1$, common successors, already provided by totality, make the colimit in groups of the diagram M abelian. The fundamental group presentation of $X_M = \mathrm{hocolim}_A BM$ therefore identifies its fundamental group canonically with G; the transport part of that presentation is killed by $\pi_1(BA) = 0$. The assumed 1-truncation makes $X_M \to BG$ a weak equivalence. In the homology spectral sequence, $E^2_{1,1} = L_1 \operatorname{colim}_A M$: the only potentially incoming differential is from $H_3(BA) = 0$. The edge map on second homology is the canonical

$$\operatorname*{colim}_A \Lambda^2 M \longrightarrow \Lambda^2 G.$$

It is surjective because two representatives can be moved to a common successor. Since $H_2(X_M) \to H_2(BG)$ is an isomorphism, the edge filtration exhausts $H_2(X_M)$ and $L_1 \operatorname{colim}_A M = 0$. Exactness follows. Now the edge map is an isomorphism; the natural decomposition $\Lambda^2(M \oplus N) = \Lambda^2 M \oplus (M \otimes N) \oplus \Lambda^2 N$ gives the same tensor comparison and siftedness. This checks the argument credited in the supplied prompt to PRETOT F.6 as well as TOTEX D.5. For the converse, filtered homotopy colimits preserve n-truncation: the tests for homotopy groups and for equality of their elements involve finite simplicial data. □

Remark D.3 (Known: the positive-truncation result retained from R6)**.** For an arbitrary small category A, including the empty and disconnected cases, put $\ell : A \to \Pi A$ for groupoid completion. For each $a \in A$ let

$$C_a = \mathrm{El}_A(\mathrm{Hom}_{\Pi A}(a, \ell(-))).$$

Thus an object is $(x, \gamma : a \to \ell x)$, and a morphism $(x, \gamma) \to (y, \delta)$ is $f : x \to y$ in A with $\ell(f)\gamma = \delta$. The checked R6 theorem says, for every fixed finite $n \geq 1$, that homotopy colimits over A preserve n-types if and only if all these categories C_a are filtered. We re-read the supplied proof in TOTEX D.9–D.10, including its simply connected case and the universal-cover comparison. The covering comparison is the pullback of $BC_a \to BA$ along $\operatorname{hocolim}_A X \to BA$, modeled by $\operatorname{hocolim}_{C_a} X|_{C_a}$: over a bar simplex a lift is a choice of the initial path class γ, and determines all remaining lifts. This specifies the comparison used, rather than treating passage to a cover as an automatic transfer principle.

In particular all the positive finite truncation classes coincide. Their intersection with componentwise totality is componentwise filteredness. On nonempty connected categories a separator between totality and exactness would therefore be nonsifted and would preserve no positive finite truncation level. These exclusions were already established in R6; the present manuscript audit does not claim them as new results.

Proposition D.4 (Known corrections: the two types and XII's exact quantifiers)**.** *The following are the corrected readings of the supplied edition.*

(i) At II ms. 36, for $f : X \to Y$, the condition printed as connectedness of $y\backslash Y$ must be connectedness of $y\backslash X = (y \downarrow f)$, for each $y \in Y$.

(ii) At XII ms. 88 the displayed formula must be $U'/t' \simeq U'/H'$, with $H' = p_! H$, rather than U/H'.

(iii) XII Definition 1(a), ms. 88–89, concerns a presheaf $H : U^{\mathrm{op}} \to \mathrm{Set}$ and all functors $p : U \to U'$. It requires the presheaf $p_! H : (U')^{\mathrm{op}} \to \mathrm{Set}$ to have W-aspherical category of elements for every such p. Definition 1(b) applies this definition to the terminal presheaf on U. It is this category totality, with $W = W_\infty$, that is compared with exactness here.

Proof. The category $y\backslash Y$ always has initial object id_y; its connectedness cannot express finality of f. In contrast, $(y \downarrow f)$ has objects $(x, \alpha : y \to f(x))$ and arrows $r : x \to x'$ with $f(r)\alpha = \alpha'$. Nonemptiness and connectedness of all these comma categories are precisely ordinary finality. This is the typed form required by the question on II ms. 36. Under the R1 equivalences supplied in the prompt, totality says that every ordinary final image has weakly contractible classifying space; it is not merely the finality of one particular map.

For XII we keep the presheaf variance explicit. Write $\pi_H : U/H \to U$ for the category with objects (u, h), $h \in H(u)$, and arrows $r : (u, h) \to (v, k)$ satisfying $H(r)k = h$. The adjunction

$$p_! : \mathrm{Set}^{U^{\mathrm{op}}} \rightleftarrows \mathrm{Set}^{(U')^{\mathrm{op}}} : p^*$$

has $p^* F' = F' \circ p^{\mathrm{op}}$ and the pointwise formula

$$(p_! H)(u') = \pi_0(u' \downarrow (p \circ \pi_H)).$$

This types the edition's formula $\pi_0(u'\backslash U/H)$. Since H' is a presheaf on U', only U'/H' is its category of elements. Taking $H = e_U$ gives $(p_! e_U)(u') = \pi_0(u' \downarrow p)$, exactly Definition 1(b). In particular there is a canonical isomorphism $(U/H)^{\mathrm{op}} \cong \mathrm{El}_{U^{\mathrm{op}}} H$, with H on the right treated as a covariant functor on U^{op}. This explicitly relates the two variance conventions.

The presheaf assertion in Definition 1(a) quantifies over extensions along maps out of its given base U. It does not, by that wording alone, quantify over all actions on U/H. The supplied PRETOT presheaf counterexample prevents replacing that restricted statement by category totality of U/H. Consequently neither the corrected typography nor that presheaf formulation supplies the missing total-to-exact implication. The two open adjacent comparisons remain exactly those recorded in Proposition D.1. □

Remark D.5 (Known: scope of the literature record)**.** The following audit records the primary texts and exact passages opened in this session, together with unsuccessful retrieval routes for the requested texts that were not obtained. It does not infer an unpublished mathematical conclusion from the absence of a search result.

D.2 Primary-source audit and main-question search

The following records the actual reading in this session. "Not obtained" means that bibliographic records were opened but the mathematical body was not available through the routes tried. No theorem number or proof is attributed to an unopened body.

Source	Opened passages or unsuccessful attempts	Bearing on this record
Isbell and Mitchell (1973)	Full supplied article, all three pages: Theorem 1, Lemma 1, Theorem 2, Corollaries 1–2, Theorem 3.	Exactness criterion and finite-family reduction; weak-terminal exclusion; announcement of the increasing-injection example.
Isbell and Mitchell (1976)	Full supplied article: Proposition 1.1; Lemmas 2.1–2.2; Proposition 2.3; Theorem 3.1; Lemma 3.2; Proposition 3.3; coefficient remark on p. 296; Theorem 4.1; Corollaries 4.2–4.4; Proposition 4.5; Corollary 4.6; note added in proof.	Purity and affine filteredness; I+III; transport of IV to connected set diagrams; endomorphism, weak-terminal and bounded-hom exclusions. The note rejects the claimed implication I+V⇒III in the 1975 paper.
Isbell (1968)	Full Cambridge PDF, four pages. Both unnumbered theorems: §1, p. 569, and §2, p. 571.	A finite-family exactness criterion and the one-object case. The minimal-period argument was checked directly. The original has no numbered "Theorem 1" or "Theorem 2".
Isbell (1974)	Full text not obtained. Official Annals article page and JSTOR record opened; JSTOR PDF route returned HTML. The Annals page supplied only general PDF-access information.	Bibliography and the 1976 article's attribution only. No original proof or numbered theorem claimed as read.
Oberst (1968)	Full text not obtained. Springer article page opened; its PDF link returned the subscription article page. Exact-title, GDZ and EuDML searches did not locate an accessible full text.	No unverified theorem number used. Husainov's separately opened Theorem 2.7 is distinguished from Oberst's original article.
Gabriel and Ulmer (1971)	Full PDF from Douglas Ravenel's academic archive. §15.1; Example 15.2(c), p. 187, and its proof in §15.9, pp. 190–191; also §§14.8–14.14 and §8.9.	The original finite-cocone connectedness criterion for siftedness. The proof in §15.9 is needed because Example 15.2(c) announces it before proving it. No total-to-exact theorem was identified in these passages.
Adámek and Reiterman (1975)	Full text not obtained. The AMS PDF, article page and DOI endpoint returned HTTP 403. Two ResearchGate article records were opened and had no full text.	Bibliography only; no proof content inferred. The criticism in Isbell–Mitchell 1976 is attributed to that opened source.

Source	Opened passages or unsuccessful attempts	Bearing on this record
Adámek et al. (1981)	Supplied complete scan, source images and OCR. Construction B.1, B.4–B.5, C.1, and the full C.2 proof I–VI, including V.2.1–V.2.3, were personally checked by the topology agent under the repaired conditions (9)–(11); details and index corrections are in Part C.	Quasi-filtered construction, incidence inequalities and the non-affine-equalizable triple. DML metadata's first-name error is avoided: the third author is Jan Reiterman.
Husainov (2002)	Full author-posted text opened on ResearchGate; §§2.1, 2.3, Theorems 2.5–2.7, Problem 2.5 and corresponding references inspected.	Affine filteredness, linearization of injective-action diagrams, and exactness in AB5 abelian categories. Problem 2.5 is not treated as evidence of present unresolved status.
Sedaghatjoo and Khaksari (2017)	Complete arXiv preprint and published PDF; Propositions 2.1, 2.3; Theorem 3.2; Lemma 3.7; Corollary 3.8; Theorem 3.9; Proposition 3.12.	For the one-element right act, all-product flatness, super-flatness and existence of a left zero are equivalent (Proposition 3.12). Finite-product flatness is separately treated in Theorem 3.2. These are ordinary set tensor products.
Laan et al. (2021)	Complete primary PDF; Definitions 1.1–1.3, Remark 1.6, Theorem 3.5, Theorem 4.2, Proposition 4.4 and Proposition 5.9.	Distinguishes pullback, finite-limit and all-limit flatness; conditions (P), (E) and local cyclicity. No inference from permutation module acyclicity to flatness of $\Delta\mathbb{Z}$ is supplied.
Bulman-Fleming and Laan (2001)	Full text not obtained. Author publication list and its public publisher share link opened; the share link yielded no readable text.	A foundational act-flatness reference cited by the two opened papers above; no original theorem quoted as read.

The proof checks needed for our uses are as follows. In the proof of Isbell–Mitchell's Theorem 3.1, the coefficients produced by purity can initially have different targets. Condition I transports their finite support to a common target; the objectwise equations remain zero. Proposition 3.3 then uses the union of the supports of two affine arrows and condition III, rather than applying IV to affine arrows. In Theorem 4.1 the periods belong to an endomorphism acting on a set; no invertibility is assumed. The weakly terminal object is used later in Corollary 4.4. In Gabriel–Ulmer's terminology, "kofinal" is final for our covariant diagrams and "kofiltrierend" is filtered. Their two-object cocone criterion is therefore finality of the diagonal, which is siftedness and does not assert totality.

For the main implication we searched variants of "exact colimits" with "aspherical", "totally aspherical", "homotopical", "Isbell", "permutation modules" and "homotopy flat"; also "homologically flat acts", "linearization of acts", "Tor permutation modules flat monoid", "homological dimension atomic topos", and "smooth coinvariants oligomorphic". We used both search engines and opened primary PDFs where found. Searches under "totally aspherical" also returned the different test-category terminology about aspherical diagonals, which we did not substitute for the definition in this problem.

The closest exact question found is Henry (2026), posted on 22 April 2026. It asks which shapes have all set colimits preserved by Set $\hookrightarrow$ Spaces; the comments identify the condition with weakly contractible components of all categories of elements. No answer was visible on the opened page. This is a retrieval record, not evidence that the present main question has a particular mathematical status.

Further opened primary sources were Stanculescu (nd), Definition 1.1, Propositions 1.2 and 1.7, and Definition 2.8, on hocolimit-commutation classes; Caramello (2013, Theorem 3.5), on the atomic-topos/Fraïssé comparison; and Harman and Snowden (2024), §§1.3.3–1.3.4, 10–11, on measure-dependent representation categories. The latter use Schwartz spaces and a completed algebra depending on a measure; their semisimplicity cannot be transferred to ordinary smooth G-modules without an additional comparison theorem. The Ramsey sources used in our sufficient condition are Mašulović (2026, Definition 2.1, Example 2.1 and §4), Hubička and Nešetřil (2019, Theorem 3.21 and Corollaries 4.2–4.3), and Hubička (2025, Theorem 15 and its proof). No novelty is claimed.

Retrieval programs and their logs are included under `code/`. The audit records unsuccessful requests as well as successful ones.

References

Adámek, J., Koubek, V., and Reiterman, J. (1981). Embeddings into categories with fixed points in representations. *Czechoslovak Mathematical Journal*, 31(3):368–389. Opened: attached primary scan, images and OCR; passages and exact extent of the construction-induction check recorded in Part C.

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Grothendieck, A. (2001). Dérivateurs: Chapters II and XII. Künzer–Malgoire–Maltsiniotis edition; supplied transcription. Opened: Chapter II, manuscript pages 36–36ter (edition dated 8 July 2001), and Chapter XII, manuscript pages 88–89 (edition dated 7 December 2002). The year identifies the displayed edition, not the original manuscript date.

Harman, N. and Snowden, A. (2024). Oligomorphic groups and tensor categories. Opened: full primary preprint; Sections 1.3.3–1.3.4, 10 and 11 inspected. These are measure-dependent Schwartz-space representations, not an asserted comparison with ordinary smooth coinvariants.

Henry, S. (2026). Which colimit shape are preserved by Sets → Spaces. MathOverflow question, 22 April 2026. Opened: question and comments; no answer visible on 13 November 2026. Used only as a search record, not a proof or unresolved-status authority.

Hubička, J. (2025). Big Ramsey degrees using parameter spaces. Opened: full primary preprint; Theorem 15 and the complete Section 5.1 proof of the finite ordered triangle-free graph Ramsey theorem.

Hubička, J. and Nešetřil, J. (2019). All those Ramsey classes (Ramsey classes with closures and forbidden homomorphisms). *Advances in Mathematics*, 356:106791. Opened: full primary preprint; Theorem 3.21, Definition 4.1, Corollaries 4.2–4.3 and their proofs inspected.

Husainov, A. A. (2002). Homological dimension theory of small categories. *Journal of Mathematical Sciences*, 110(1):2273–2321. Opened: full author-posted text; Sections 2.1 and 2.3, Theorems 2.5–2.7 and Problem 2.5 inspected.

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